

FMCG AI SUPPLY CHAIN MANAGEMENT SYSTEM

Product Requirements Report (PRD)

Version: 1.0 — Frozen Baseline

Date: September 2026

Product Type: AI-enabled FMCG Supply Chain Decision Support & Control System

1. Executive Summary

The FMCG AI Supply Chain Management System is an event-driven, AI-assisted supply-chain decision platform designed to reduce the impact of operational disruptions across Procurement, Inventory, Production, and Logistics.

The core business problem is not simply a lack of data. FMCG organizations often have operational information distributed across departments, causing delays in recognizing how one disruption affects other functions. A supplier delay, for example, may initially appear to be a procurement issue but can subsequently create inventory shortages, production disruption, customer-service risk, and logistics consequences.

The system addresses this problem by converting operational events into a coordinated decision process:

Event → Detection → Targeted Routing → Domain Analysis → Risk → Impact → Scenarios → Decision → Approval → Execution → Verification → Risk Recalculation

The system contains:

- 4 operational domains
- 4 domain AI agents
- 1 Master Agent / Master Orchestrator
- 4 role-specific frontend dashboards
- 5 SNS Workbench workflows
- Central Risk, Impact, Decision, Forecasting, and Replenishment intelligence
- Human-in-the-loop approval
- Closed-loop verification

- **Supabase PostgreSQL as the operational source of truth**

The product is designed to be practical, explainable, laptop-friendly, and implementable without unnecessary distributed infrastructure.

2. Core Business Problem

2.1 Primary Pain Point

The core pain point is:

Supply-chain disruptions are often handled department-by-department, making cross-functional impact difficult to identify quickly and causing reactive rather than proactive decisions.

Typical flow:

Supplier Delay → Procurement notices → Inventory is informed → Production reacts → Logistics reacts → Management intervenes

This creates:

- **Delayed response**
- **Fragmented information**
- **Manual coordination**
- **Poor visibility of downstream consequences**
- **Excessive communication**
- **Stockout and production risks**
- **Difficulty selecting the best response**
- **Limited traceability of decisions**
- **No reliable verification of whether an action solved the problem**

2.2 Product Solution

The system creates a coordinated intelligence layer over existing operational processes.

Example:

Supplier Delay

- Procurement analyzes supplier impact
 - Inventory analyzes stock exposure
 - Production analyzes material/production exposure
 - Master Agent combines findings
 - Risks and scenarios are evaluated
 - Best action is recommended
 - Human approval is obtained when required
 - Action is executed
 - Actual result is verified
 - Risk is recalculated
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3. Product Objectives

The product shall:

1. Detect important operational supply-chain events.
 2. Identify which domains are actually affected.
 3. Route events only to relevant domain agents.
 4. Analyze domain-specific risks.
 5. Identify forward and backward impacts.
 6. Generate and evaluate response scenarios.
 7. Produce explainable recommendations.
 8. Coordinate multiple domain agents through a Master Agent.
 9. Require human approval for defined high-impact actions.
 10. Execute approved actions through controlled backend services.
 11. Verify actual outcomes against expected outcomes.
 12. Recalculate risk after execution.
 13. Maintain an auditable AI decision trail.
 14. Provide each department with its own authorized operational dashboard.
 15. Maintain one operational source of truth.
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4. Product Architecture

4.1 Four Operational Domains

The system has exactly four primary operational stages:

1. Procurement
2. Inventory
3. Production
4. Logistics

4.2 Four Domain Agents

Each domain has exactly one primary AI agent:

Domain	AI Agent	Primary Responsibility
Procurement	Procurement Agent	Suppliers, POs, procurement risks
Inventory	Inventory Agent	Stock, safety stock, stockout/overstock
Production	Production Agent	Material availability, production impact
Logistics	Logistics Agent	Shipments, delivery and transportation risk

4.3 Master Agent

There is exactly one:

Master Agent / Master Orchestrator

It coordinates the four domain agents and performs cross-domain reasoning.

It is not a fifth operational department.

5. Frontend Product Requirements

5.1 Four Role-Specific Dashboards

The product shall provide four operational dashboards:

Procurement Dashboard

Displays:

- Supplier status
- Purchase orders
- Supplier delays
- Lead-time risk
- Procurement recommendations
- Procurement alerts
- Relevant decisions and approvals

Inventory Dashboard

Displays:

- Total stock
- Reserved stock
- Available stock
- Safety stock
- Stockout risk
- Overstock risk
- Replenishment requirements
- Inventory alerts

Available Stock = Total Stock – Reserved Stock

Production Dashboard

Displays:

- **Production plans**
- **Production orders**
- **Material availability**
- **Production constraints**
- **Schedule risks**
- **Production impact**
- **Production recommendations**

Logistics Dashboard

Displays:

- **Shipments**
- **Shipment status**
- **Delivery status**
- **Transportation constraints**
- **Delivery risks**
- **Logistics recommendations**

5.2 Login and Access Control

There shall be one common authentication system.

Flow:

Login → Authentication → User Identity → Role Resolution → Permission Check → Authorized Dashboard

Example:

Procurement User → Procurement Dashboard

A procurement user must not access Inventory, Production, or Logistics dashboards without authorization.

Security must be enforced at:

- **Frontend route level**
- **FastAPI backend**

- Application RBAC
- Supabase Auth
- Supabase Row Level Security (RLS)

Frontend hiding alone is not considered sufficient authorization.

6. SNS Workbench Requirements

SNS Workbench shall contain exactly five major workflows:

Workflow 1 — Procurement Agent

Workflow 2 — Inventory Agent

Workflow 3 — Production Agent

Workflow 4 — Logistics Agent

Workflow 5 — Master Agent / Master Orchestrator

These workflows represent backend intelligence, not separate applications.

7. Multi-Agent Event Processing

7.1 Core Event Architecture

The standard workflow is:

Event Producer

→ **Event Store/Event Bus**

→ **Event Router**

→ **Affected Domain Identification**

→ **Relevant Agent(s)**

→ **Risk Detection**

→ **Impact Analysis**

→ **Scenario Generation**

- Scenario Evaluation
- Decision Optimization
- Master Agent
- Recommendation
- Approval
- Controlled Execution
- Verification
- Risk Recalculation
- Continuous Monitoring

7.2 Event Ownership vs Event Impact

An event can originate from one domain but affect multiple domains.

Therefore the system must distinguish:

origin_domain — where the event originated

from:

affected_domains — which domains need to analyze it.

Example:

SupplierDelayDetected

origin_domain = PROCUREMENT

affected_domains =

PROCUREMENT

INVENTORY

PRODUCTION

Logistics is added only if dependency analysis determines that it is affected.

8. Cross-Domain Event Example

Milk Supplier Delay

Example event:

Event ID: EVT-001

Event Type: SupplierDelayDetected

Supplier: Supplier A

Material: Milk

Expected Delivery: 2026-09-06

New Delivery: 2026-09-09

Delay: 3 days

Priority: HIGH

Processing:

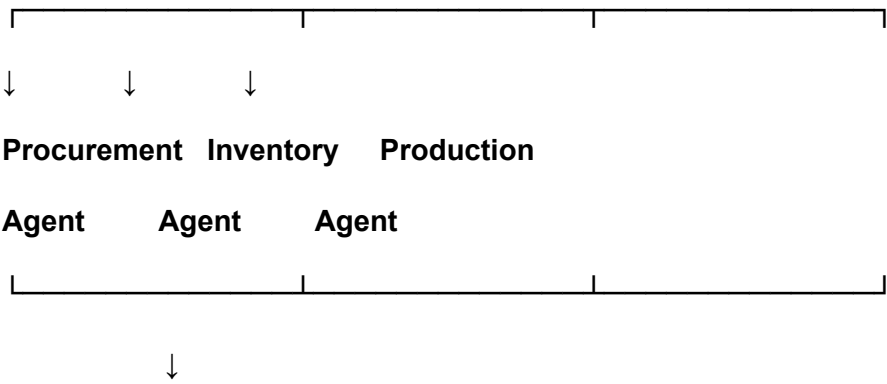
Supplier Delay



Event Validation



Event Router



Master Agent



Cross-domain analysis



Risk assessment



Scenario evaluation



Recommendation

Procurement Agent

Checks:

- **Supplier**
- **Purchase order**
- **Lead time**
- **Supplier performance**
- **Expedite possibilities**
- **Alternative suppliers**

Inventory Agent

Checks:

- **Current stock**
- **Reserved stock**
- **Available stock**
- **Safety stock**
- **Consumption**
- **Projected stockout**

Production Agent

Checks:

- **Material requirements**
- **Production orders**
- **Production schedule**
- **Material shortage impact**

Master Agent

Combines all findings and determines the coordinated response.

Important: The agents do not independently make conflicting final decisions. They provide domain analysis and recommendations to the Master Agent.

9. Targeted Routing Requirement

The system shall NOT blindly broadcast every event to all four agents.

Incorrect:

Every Event → Procurement + Inventory + Production + Logistics

Correct:

Event → Dependency Analysis → Relevant Agents

For example:

Supplier Delay

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Procurement affected

↓

Inventory affected

↓

Production affected

↓

Check Logistics dependency



Invoke Logistics only if affected

This reduces unnecessary processing, alert noise, and agent workload.

10. Central Intelligence

The following are centralized intelligence services, not additional agents:

Risk Detection Engine

Identifies and scores operational risks.

Impact Analysis Engine

Determines forward and backward consequences.

Decision Optimization Engine

Evaluates alternatives and recommends preferred actions.

Demand Forecasting

A lightweight Python ML pipeline used for demand prediction.

Smart Replenishment

A deterministic service/engine that determines replenishment requirements.

No unnecessary additional AI agents should be introduced.

11. Human-in-the-Loop

The AI system is a decision-support and controlled-action system.

AI may:

- Detect
- Analyze
- Forecast
- Generate scenarios
- Recommend

AI must not have unrestricted authority over critical business operations.

Controlled flow:

AI Recommendation → Approval Policy → Human Approval → Execution

Human approval should be required where appropriate for:

- High-value procurement
 - Supplier switching
 - Major production changes
 - Significant inventory movements
 - Financially significant actions
 - Other business-critical operations
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12. Closed-Loop Verification

A recommendation is not considered successful merely because an action was executed.

Required process:

Action → Execution → Actual Result → Expected vs Actual Comparison → Verification → State Update → Risk Recalculation

If the issue remains unresolved:

Reassess → Impact Analysis → Scenario Generation → Recommendation → Approval → Execution → Verification

This allows the system to continuously respond to changing conditions.

13. Data and System of Record

Supabase PostgreSQL shall remain the single operational source of truth.

Operational entities include:

- Products
- Suppliers
- Warehouses
- Inventory
- Batches
- Stock movements
- Purchase orders
- Production plans
- Production orders
- Production materials
- Shipments
- Demand records
- Forecasts
- Events
- Dependencies
- Risks
- Decisions
- Recommendations
- Approvals
- Actions
- Executions
- Verification results
- Alerts
- Notifications
- Reports
- Audit logs

AI agents must not create conflicting operational copies of these entities.

Agent memory is limited to temporary reasoning/context.

14. AI Traceability

Every significant AI decision should be traceable through:

Event → Affected Entity → Risk → Impact → Scenario → Recommendation → Approval
→ Execution → Verification

This supports:

- Explainability
 - Debugging
 - Auditing
 - Accountability
 - Performance evaluation
 - Human review
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15. Technology Requirements

Layer	Technology
Frontend	React
UI Framework	Tailwind CSS
Charts	Recharts
UI Design	Google Stitch
Development	Antigravity
Backend	Python + FastAPI

Database **Supabase PostgreSQL**

Authentication **Supabase Auth**

Authorization **Supabase RLS + FastAPI
RBAC**

AI Agents **SNS Agent Workbench**

Forecasting **Lightweight Python ML**

**Version
Control** **Git + GitHub**

Architecture pattern:

**Modular Monolith + Event-Driven Intelligence Layer + Multi-Agent Domain Reasoning +
Human-in-the-Loop + Closed-Loop Verification**

The system should avoid unnecessary:

- **Kubernetes**
 - **Microservices**
 - **Large distributed infrastructure**
 - **Deep-learning forecasting**
 - **Large custom model training**
 - **Excessive agents**
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16. MVP Requirements

The first working MVP should demonstrate one complete disruption scenario.

Primary MVP Scenario

Milk Supplier Delay

The system must demonstrate:

1. Event creation
2. Event validation
3. Targeted routing
4. Procurement analysis
5. Inventory analysis
6. Production analysis
7. Master Agent coordination
8. Risk detection
9. Impact analysis
10. Scenario generation
11. Decision recommendation
12. Human approval
13. Controlled execution
14. Verification
15. Risk recalculation
16. Updated dashboard state

The system should demonstrate that a single event can affect multiple domains without requiring all agents to be invoked unnecessarily.

17. Success Criteria

The product will be considered successful when it can demonstrate:

Visibility

A disruption is visible to all relevant departments.

Targeted Intelligence

Only affected domain agents are invoked.

Cross-Functional Reasoning

The Master Agent combines multiple domain analyses.

Decision Quality

The system produces understandable alternative scenarios and recommendations.

Governance

Important decisions pass through appropriate approval controls.

Execution

Approved actions are executed through controlled backend services.

Verification

The system confirms whether the expected outcome actually occurred.

Traceability

The complete decision chain can be audited.

Security

Users can access only the dashboards and data authorized for their roles.

Data Integrity

Supabase remains the authoritative operational state.

18. Frozen Product Principles

The following principles are now the baseline for implementation:

1. Exactly four operational domains.
 2. Exactly four primary domain agents.
 3. Exactly one Master Agent / Master Orchestrator.
 4. Exactly four primary user dashboards.
 5. Exactly five SNS Workbench workflows.
 6. One event may affect multiple domains.
 7. Event origin and affected domains are separate concepts.
 8. Routing is dependency-based and targeted.
 9. Domain agents analyze and recommend.
 10. The Master Agent coordinates cross-domain decisions.
 11. Central intelligence remains logically separate from domain agents.
 12. Human approval controls high-impact actions.
 13. Execution is controlled through backend services.
 14. Verification is mandatory after execution.
 15. Risk is recalculated after verification.
 16. Supabase PostgreSQL is the operational source of truth.
 17. AI reasoning must be traceable.
 18. Frontend authorization must be backed by server-side authorization.
 19. The system must remain practical and laptop-friendly.
 20. No unnecessary agents or infrastructure should be introduced.
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19. Final Product Definition

The FMCG AI Supply Chain Management System is a multi-agent, event-driven supply-chain decision and control platform in which four specialized domain agents analyze operational disruptions and one Master Agent coordinates cross-domain decisions.

Its core value is not simply the use of AI agents.

Its core value is the ability to transform:

Operational Event

into:

Targeted Analysis → Risk Understanding → Cross-Domain Impact → Scenario Evaluation
→ Coordinated Decision → Controlled Action → Verified Outcome

The product therefore addresses the fundamental FMCG supply-chain problem of fragmented, delayed, reactive decision-making by creating a coordinated, explainable, and closed-loop operational intelligence system.

Frozen architecture baseline:

4 Dashboards + 4 Domain Agents + 1 Master Agent + 5 SNS Workflows + Central Intelligence + Human Approval + Controlled Execution + Verification + Continuous Monitoring.